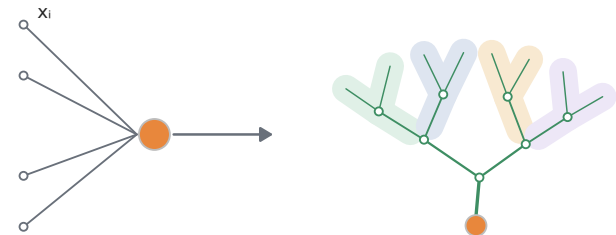
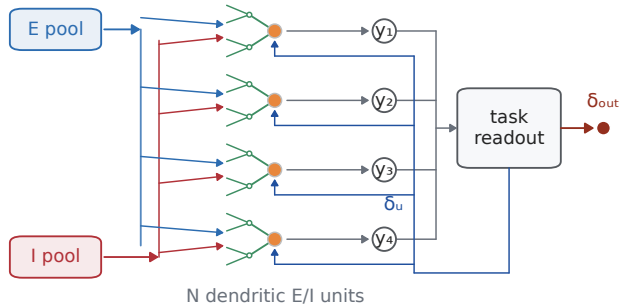
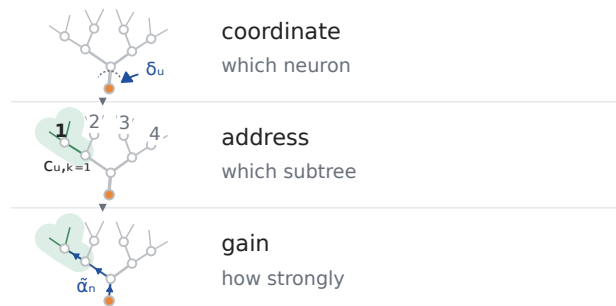
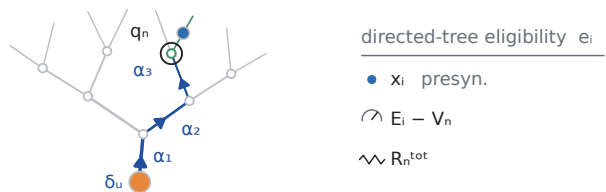


APoint unit $\rightarrow$ dendritic treeone neuronal coordinate δ_u route signals $c_{u,k}$ **B**

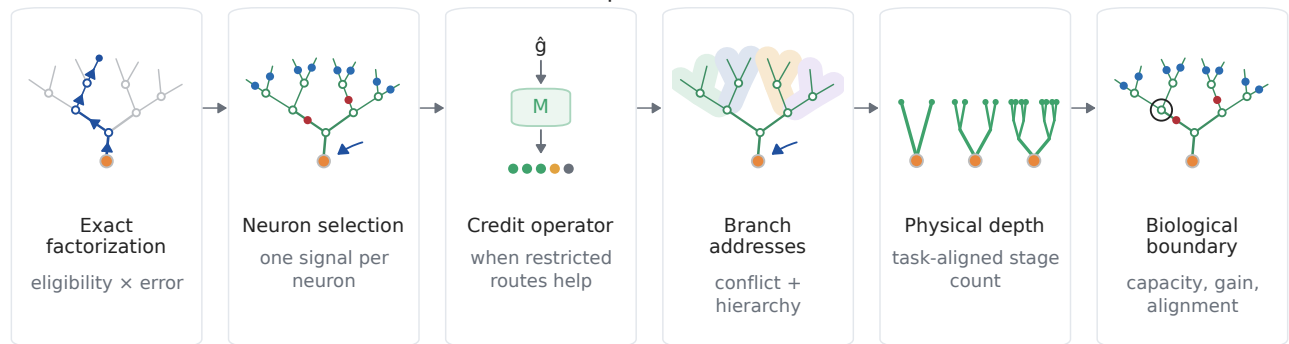
Network layer



N dendritic E/I units

CCoordinate $\rightarrow$ address $\rightarrow$ gaincoordinate
which neuronaddress
which subtreegain
how strongly**D**Local eligibility $\times$ transported errordirected-tree eligibility e_i $\bullet$ x_i presyn. $\bigcirc$ $E_i - V_n$ $\sim$ R_n^{tot} directed tree: $\partial \mathcal{L} / \partial g_i = [x_i R_n^{\text{tot}} (E_i - V_n)] [\partial \mathcal{L} / \partial V_n]$ general adjoint: $\partial \mathcal{L} / \partial g_i = x_i (E_i - V_n) q_n$, $q_n = R_n^{\text{tot}} \partial \mathcal{L} / \partial V_n$ **E**

Roadmap of the Results

Exact
factorization
eligibility $\times$ errorNeuron selection
one signal per
neuronCredit operator
when restricted
routes helpBranch
addresses
conflict +
hierarchyPhysical depth
task-aligned stage
countBiological
boundary
capacity, gain,
alignment